SOPHIA J. HOLLICK

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EDUCATION

Yale University *March 2025*
PhD Physics
Advisor: Dr. Reina H. Maruyama

Yale University *December 2023*
Masters of Science Physics
Masters of Philosophy Physics
Advisor: Dr. Reina H. Maruyama

Rensselaer Polytechnic Institute *May 2020*
B.S. Physics *Summa Cum Laude* GPA: 3.98/4.0

Marshwood High School *June 2017*
High Honors GPA: 4.1/4.0
South Berwick, ME

RESEARCH EXPERIENCE

Visiting Graduate Student Researcher Universidad de Zaragoza
COSINE-100 and ANAIS-112: Combined Dark Matter Search *September 2023–April 2025*

- ANAIS-112, located in Canfranc, Spain, seeks to perform a model-independent test of the DAMA/LIBRA dark matter claim and is the sister experiment to COSINE-100. I am working in Spain to facilitate a combined search for dark matter between the two experiments.
- Additionally, I conducted analysis studies on the ANAIS-112 background model and found that a substantial difference in fine-tuning between performing fits with ROOfit and Markov-Chain Monte Carlo (MCMC) can be realized. Due to difficulty modeling the low energy region, the MCMC is useful for understanding which particular components are behaving poorly.

Particle Physics Graduate Student Researcher Yale University
COSINE-100: Dark Matter Direct Detection *June 2021–May 2025*

- COSINE-100 is a collaboration between the DM-Ice and KIMS experiments where NaI(Tl) detectors are used in order to perform a model-independent test of the DAMA/LIBRA experiment who claim to observe a dark matter signal taking the shape of an annual modulation as expected from standard halo model dark matter.
- My thesis focused on the combined search for dark matter with the COSINE-100 and ANAIS-112, a sister experiment also utilizing NaI(Tl) to test DAMA, experiments. Through this combined analysis, results consistent with no modulation, excluding the DAMA signal by $\geq 3\sigma$, were observed.
- Efforts for optimizing radiopurity in next generation NaI(Tl) detectors by measuring their cosmogenic activation rates with artificially activated NaI(Tl) crystals were investigated to understand their impact on the dark matter search.
- A replication of the DAMA/LIBRA background subtraction procedure was performed on COSINE-100 data and on simulated DAMA/LIBRA data (from COSINE-100 known backgrounds) to show analysis artifacts influencing the final modulation signal as resulting from time-dependent backgrounds.

- Varying geometries for the proposed COSINE-200 experiment were simulated in GEANT4 to study the variations in the NaI(Tl) crystal backgrounds. Crystals with smaller surface areas were observed to yield less active backgrounds as ^{210}Pb is a common contaminant on the crystal surface.

Nuclear Astrophysics Intern — Brookhaven National Laboratory
Nuclear Data Analysis — *Fall 2019 and Summer 2020*

- Developed Decision Tree Machine Learning algorithms with the National Nuclear Data Center (NNDC).
- Developed Python program for determining spacing of nuclear level densities with reliable uncertainty of any given nucleus in The Atlas of Neutron Resonances by testing a variety of mathematical models on the structure of the nucleus' cumulative level distribution.
- Computed D-spacings by l and j spin groups

Plasma Physics Research Assistant — University of New Hampshire
Solar Wind Data Analysis from Voyager I and II Spacecrafts — *Spring 2016 – Summer 2018*

- Created a library of pick-up ion wave events in the form of a detailed database and wrote a series of programs to extract information about the solar wind environment, position in space, and event duration timescale.

INSTRUMENTATION AND ANALYSIS EXPERIENCE

Hamamatsu Photomultiplier Tubes (PMT)
NaI(Tl) Crystal Scintillator Detectors
CAEN Digitizers
Data Analysis with ROOT framework
GEANT4 Simulations
Machine Learning with Decision Tree Techniques

TEACHING AND MENTORING EXPERIENCE

Advanced Physics Laboratory (PHYS 382L) — Fall 2021 - Spring 2023
Teaching Fellow, Department of Physics, Yale University

Introductory Physics Laboratory 2 (PHYS 166L) — Spring 2021
Teaching Fellow, Department of Physics, Yale University

Introductory Physics Laboratory 1 (PHYS 165L) — Fall 2020, Fall 2022
Teaching Fellow, Department of Physics, Yale University

Star, Galaxies, and the Cosmos (ASTR 1520) — Spring 2019
Teaching Assistant, Department of Physics, Rensselaer Polytechnic Institute

Advising and Learning Assistance Center (ALAC) Tutor — Fall 2018 - Summer 2019
Rensselaer Polytechnic Institute

I-PERSIST Honors Physics Mentor — Fall 2018
Department of Physics, Rensselaer Polytechnic Institute

Research Mentor — Summer 2018
Space Science Center, University of New Hampshire
Study of solar wind and data analysis from Ulysses, Voyager, and ACE spacecraft.

GRANTS, HONORS, AND AWARDS

International Workshop for Identification of Dark Matter (IDM) **Young Scientist Support Grant** — 2024

Winner of Yale's **Leigh Page Prize** 2020
Winner of the RPI Physics Dept. **G. Howard Carragan Award** 2020
Inductee of **National Society of Physics Students** (Sigma Pi Sigma) 2020
Dean's Honor List at RPI Fall 2017 - Spring 2020
Rensselaer Medalist Recipient Fall 2016

ACADEMIC AND UNIVERSITY SERVICE

WIDG Seminar Coordinator Fall 2022 - Spring 2023
Department of Physics, Yale University

- Weak Interaction Discussion Group Talks (WIDG) is a biweekly seminar held at Yale's Wright Lab. Topics include Cosmology, Nuclear, Particle, Neutrino, AMO, and Theoretical Physics. The seminar is open to all members of the Yale Physics community. Coordinators invite speakers, host the talks, and work with Wright Lab admins to make their necessary arrangements and advertise the seminars.

Graduate Student Ambassador Spring 2022
Department of Physics, Yale University

- Ambassadors help recruit new students by reaching out to accepted graduate students via email. They also assist in the open house events for the graduate physics school.

Graduate Student Advisory Committee (GSAC) 2022
Department of Physics, Yale University

- GSAC advises the Chair, the Director, and other faculty on matters related to graduate students and the graduate program, including helping plan and organize annually scheduled events. Committee members are appointed for one calendar year.

SCIENTIFIC OUTREACH

Centro de Astropartículas y Física de Altas Energías (CAPA) October 31, 2023-24
Dark Matter Day

- *"Niños y mayores están invitados a participar para aprender sobre la materia oscura realizando una serie de pruebas en la Facultad de Ciencias, incluyendo una observación astronómica con telescopios a partir de las 20:30h."* An open house day for the general public to come to the physics department at the University of Zaragoza to discover Dark Matter. Hosted in Spanish, I worked at reception, signing in guests and directing them to event locations.

Laboratorio Subterráneo Canfranc (LSC) September 24, 2023
LSC Open Day 2023

- *"The LSC organises an annual open day for individuals interested in visiting our facilities. This day is usually held on a Sunday during autumn."* Workshops, lectures, and demonstrations of detection techniques for adults and children are offered with no registration required. Open Day is held in Spanish and I assisted with the particle physics and dark matter workshop.

Girls Advancing in STEM (GAINS) 2022
GAINS Conference 2022

- *"GAINS inspires and empowers high school girls to envision, prepare for, and pursue academic and professional futures in STEM."* The GAINS conference in 2022 was hosted by Wright Laboratory at Yale. I lead an in-depth lab tour which guided the students through my work as well as the work of fellow lab mates and entertained the student's curiosity by answering their questions and sharing some results from our research studies.

PROFESSIONAL ORGANIZATIONS

CAPA Centro de Astropartículas y Física de Altas Energías 2023 – Present
SACNAS Society for the Advancement of Chicanos and Native Americans in Science 2022 – Present
American Physical Society 2020 – Present
Sigma Pi Sigma (National Physics Honors Society) Inducted 2020
Women in Physics 2017 – Present
Society of Physics Students 2017 – 2020

RELEVANT COURSES

1. **Graduate Courses**
Quantum Mechanics I&II, Advanced Theoretical Mechanics, Statistical Mechanics and Thermodynamics, Electromagnetic Theory, and Advanced Instrumentation
2. **Undergraduate Courses**
Astrophysics, Experimental Physics Laboratory, Methods of Partial Differential Equations of Mathematical Physics, and Materials Science

SKILLS

Computational skills:
ROOT, C++, Python, FORTRAN, LaTex, Jutpyter Notebook, Microsoft Office, Linux, Mac & Windows OS
Extracurriculars:
Digital Art/Design, Guitar&Piano, and Video Editing.
Languages:
English, Spanish(B2), German(A2).

TALKS & PRESENTATIONS

Invited Talks

1. ***Searching for Dark Matter with COSINE-100***
Online Seminar for Topics in Particle Physics
Tennessee, United States 8 October 2024

2. ***Searching for Dark Matter with COSINE-100***
Saturnalia 2023
`https://indico.capa.unizar.es/event/35/contributions/532`
Zaragoza, Spain 18-21 December 2023

3. ***COSINE-100 and ANAIS-112 Search for WIMPs***
WIDG Seminar at Yale
New Haven, Connecticut 28 November 2023

4. ***Searching for Dark Matter with the COSINE-100 Experiment***
20th MultiDark Workshop
`https://workshops.ift.uam-csic.es/multidarkgandia/program`
Gandía, Spain 25-27 October 2023

5. ***Searching for Dark Matter with COSINE-100***
Seminar at Universidad de Zaragoza
Zaragoza, Spain 10 October 2023

6. *Observations of Low-Frequency Magnetic Waves due to Newborn Interstellar Pickup Ions Using ACE, Ulysses, and Voyager Data* (presented by C. W. Smith)
16th Annual International Astrophysics Conference 2017
Santa Fe, CA March 2017.

Contributed Talks

1. ***"A combined search for dark matter with COSINE-100 and ANAIS-112"***
   21st MultiDark
   https://indico.ifca.es/event/3186/contributions/15754/
   Santander, Spain 9 October - 11 October 2024

2. ***"Combined analysis of COSINE-100 and ANAIS-112"***
   9th COSINE Collaboration Meeting 2024
   Chung-Ang University, Seoul, Korea 20 August - 22 August 2024

3. ***"A combined search for dark matter with COSINE-100 and ANAIS-112"***
   IDM 2024
   https://agenda.infn.it/event/39713/contributions/234125/
   L'Aquila, Italy 8 July - 12 July 2024

4. ***"Understanding the DAMA/LIBRA Signal Phase"***
   8th COSINE Collaboration Meeting 2023
   Daejeong, Korea 11 July - 14 July 2023

5. ***"Interpretation of modulating amplitude and phase from the DAMA/LIBRA analysis method"***
   (Poster)
   UCLA Dark Matter Conference 2023
   https://indico.global/event/647/contributions/16596/
   Los Angles, CA 29 March - 1 April 2023

6. ***"Cosmogenic Activation of Radioisotopes in NaI(Tl) Detectors"***
   APS April Meeting 2022
   https://meetings.aps.org/Meeting/APR22/Session/T08.3
   New York, NY 10-12 April 2022

7. *Magnetic Waves Excited by Newborn Pickup H+ Near Jupiter: Neutral Hydrogen Loss by the Planetary System*
   Charles W. Smith *et al.* 2021, *Fall Meeting of the American Geophysical Union*

8. ***"Sorting neutron resonances by spin groups using a machine learning technique."***
   2020 Fall Meeting of the APS Division of Nuclear Physics, Abstract: GA.00006,
   October 2020. (Online Poster)

9. ***"Realistic Uncertainty Determination of Nuclear Level Densities"*** (Poster)
   Early Career Research Symposium
   Brookhaven National Laboratory, November 2019

10. *"Expansion Effects in Voyager Magnetic Field Spectra"*
    Fall Meeting of the American Geophysical Union, Abstract SH23D-3364
    Washington D.C. December 2018.

11. *"Observations from 1-6 AU of Low-Frequency Magnetic Waves due to Newborn Interstellar Pickup Ions"*
    Fall Meeting of the American Geophysical Union
    San Francisco, CA December 2016.

PUBLICATIONS

1. N. Carlin et al. Combined Annual Modulation Dark Matter Search with COSINE-100 and ANAIS-112, 2025

2. G. H. Yu et al. Limits on WIMP dark matter with NaI(Tl) crystals in three years of COSINE-100 data. *in prep.*, 1 2025

3. N. Carlin et al. COSINE-100 Full Dataset Challenges the Annual Modulation Signal of DAMA/LIBRA. *in prep.*, 9 2024
4. G. H. Yu et al. Lowering threshold of NaI(Tl) scintillator to 0.7 keV in the COSINE-100 experiment. *JINST*, 19(12):P12013, 2024
5. G. H. Yu et al. Improved background modeling for dark matter search with COSINE-100. *Eur. Phys. J. C*, 85(1):32, 2025
6. S. M. Lee et al. Nonproportionality of NaI(Tl) scintillation detector for dark matter search experiments. *Eur. Phys. J. C*, 84(5):484, 2024
7. G. Adhikari et al. Alpha backgrounds in NaI(Tl) crystals of COSINE-100. *Astropart. Phys.*, 158:102945, 2024
8. G. Adhikari et al. Search for inelastic WIMP-iodine scattering with COSINE-100. *Phys. Rev. D*, 108(9):092006, 2023
9. G. Adhikari et al. Search for Boosted Dark Matter in COSINE-100. *Phys. Rev. Lett.*, 131(20):201802, 2023
10. G. Adhikari et al. Search for bosonic super-weakly interacting massive particles at COSINE-100. *Phys. Rev. D*, 108(4):L041301, 2023
11. G. Adhikari et al. Search for solar bosonic dark matter annual modulation with COSINE-100. *Phys. Rev. D*, 107(12):122004, 2023
12. R. Saldanha, W. G. Thompson, Y. Y. Zhong, L. J. Bignell, R. H. M. Tsang, S. J. Hollick, S. R. Elliott, G. J. Lane, R. H. Maruyama, and L. Yang. Cosmogenic activation of sodium iodide. *Phys. Rev. D*, 107(2):022006, 2023
13. Govinda Adhikari et al. An induced annual modulation signature in COSINE-100 data by DAMA/LIBRA's analysis method. *Sci. Rep.*, 13(1):4676, 2023
14. G. P. A. Nobre, D. A. Brown, S. J. Hollick, S. Scoville, and P. Rodríguez. Novel machine-learning method for spin classification of neutron resonances. *Phys. Rev. C*, 107(3):034612, 2023
15. G. Adhikari et al. Three-year annual modulation search with COSINE-100. *Phys. Rev. D*, 106(5):052005, 2022
16. G. Adhikari et al. Searching for low-mass dark matter via the Migdal effect in COSINE-100. *Phys. Rev. D*, 105(4):042006, 2022

OTHER WORKS

1. *Magnetic Waves Excited by Newborn Inner Source Pickup H+ Measured by the Voyager Spacecraft Inside 3 AU*
   Sophia J. Hollick *et al.* in preparation,
2. *Magnetic Waves Excited by Newborn Pickup H+ Near Jupiter: Neutral Hydrogen Loss by the Planetary System*
   Sophia J. Hollick *et al.* July 2022, *JGR Space Physics* **127**, 7 DOI
3. *Solar Wind Turbulence from 1 to 45 au. publications I. - V.*
   Zackary B. Pine *et al.* 2020, *ApJ* **900** 91, DOI *ApJ* **900** 92, DOI *ApJ* **900** 93, DOI *ApJ* **900** 94, DOI *ApJS* **250** 14, DOI
4. *Magnetic Waves Excited by Newborn Interstellar Pickup Ions Measured by the Voyager Spacecraft from 1 to 45 AU. publications I. - III.*
   Sophia J. Hollick *et al.* 2018, *ApJ* **863** 75, DOI *ApJ* **863** 76, DOI *ApJS* **237** 34, DOI

5. *Observation of Magnetic Waves Excited by Newborn Interstellar Pickup He+ Observed by the Voyager 2 Spacecraft at 30 AU*
Matthew R. Argall *et al.* 2017, *ApJ* **849** 61, DOI